PROJECT PRESENTATION

Medical Visual Question Answering for Chest X-rays

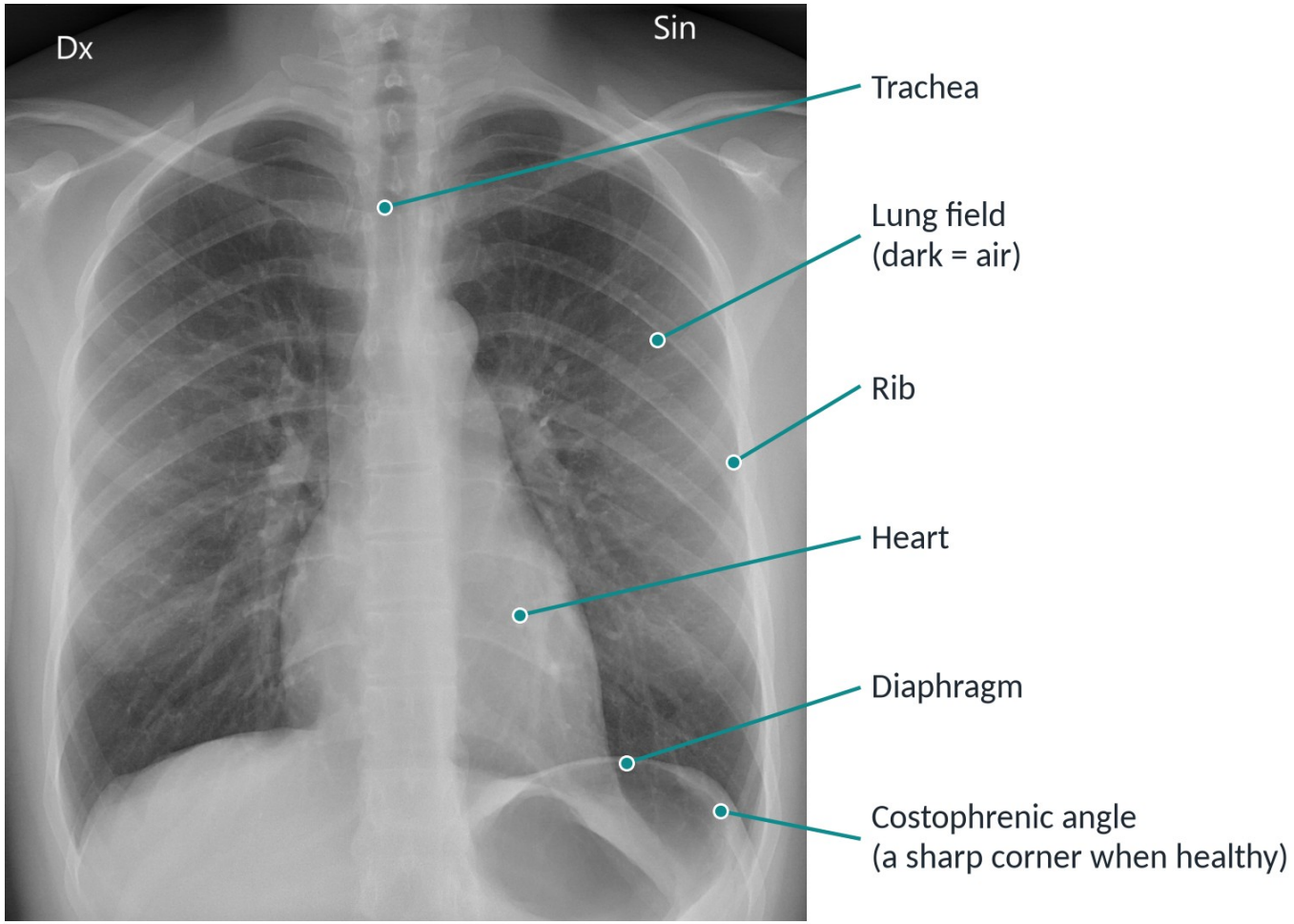
A knowledge-grounded approach with CheXagent-2-3b: literature review, baseline tests and retrieval-augmented design

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A chest X-ray is a shadow picture: dark means air, white means dense tissue



Four findings used in this project

Cardiomegaly

Heart wider than half the chest

Pleural effusion

Fluid blunts the corner at the lung base

Pneumothorax

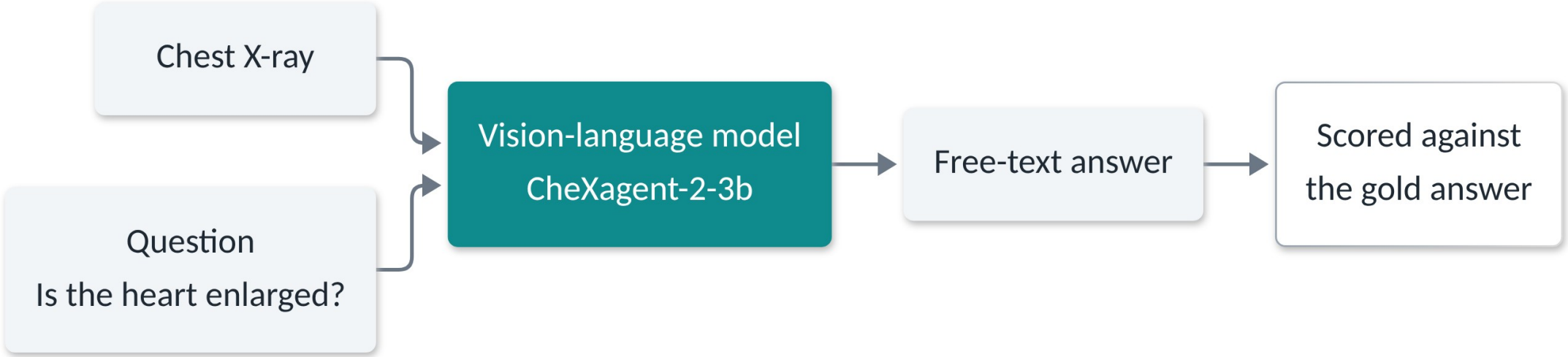
Dark area with no lung markings

Atelectasis

Collapsed lung looks whiter

Photograph: M. Haggstrom, Wikimedia Commons, CC0 (normal PA chest X-ray). Findings: standard radiology teaching material.

Medical VQA: show the model an X-ray and a question, then check its free-text answer



Perception

Can the model see the finding?

Is there an effusion? Which side? Is the heart enlarged?

Knowledge

Does the model know the disease?

What causes it? What are the symptoms? How is it treated?

Datasets: SLAKE and VQA-RAD, chest X-ray questions only. Model: StanfordAIMI/CheXagent-2-3b.

Of six papers reviewed, three shaped the project; none combines retrieval with chest X-rays

	Chest X-ray focus	Free-text answers	Uses retrieval	Clinician / blind rating	Textbook source	what we took from it
CheXagent (2024)	●	●	○	●	○	our baseline model; weakest on left/right and severity (77-79%)
RULE (2024)	○	●	●	○	○	naive retrieval hurts: 47-59% of its errors follow the text
Med-Flamingo (2023)	○	●	○	●	○	automated scores disagree with clinicians: rate blind
LLaVA-Med (2023)	○	●	○	○	○	cheap tuning; GPT-4-written training data
PMC-VQA (2023)	○	●	○	○	○	caption-made questions can be answered from text alone
PubMedCLIP (2023)	○	○	○	○	○	the image encoder matters more than the text encoder
This project	●	●	●	○	●	**none of the six papers combines these**

CheXagent is our baseline, RULE warns about retrieval, Med-Flamingo says to rate answers by hand.

Chen et al. 2024 (CheXagent); Xia et al. 2024 (RULE); Moor et al. 2023 (Med-Flamingo); Eslami et al. 2023 (PubMedCLIP); Li et al. 2023 (LLaVA-Med); Zhang et al. 2023 (PMC-VQA).

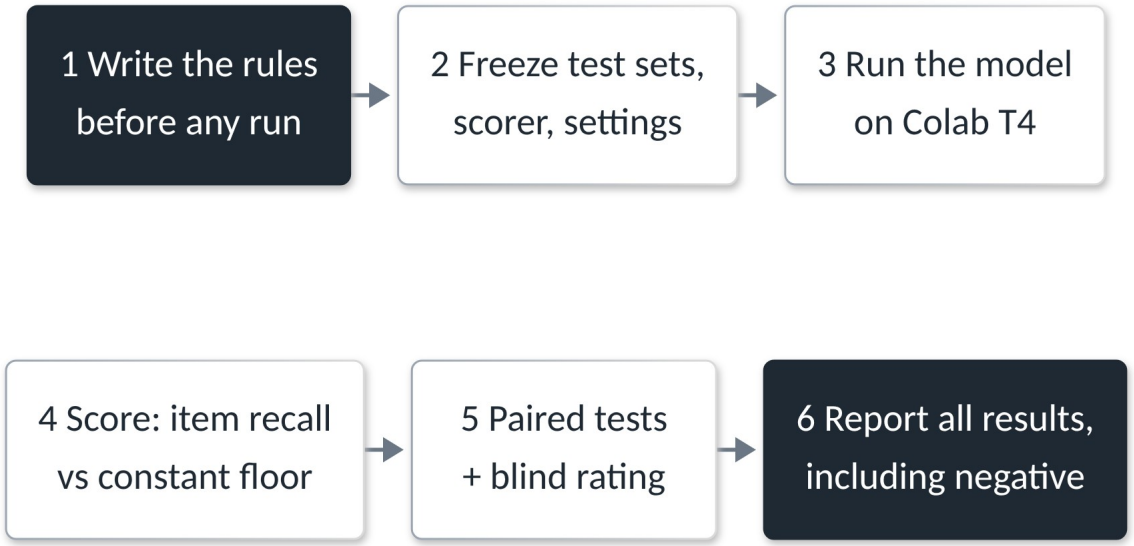
Each finding in the literature became a concrete design decision

What the papers show	What we decided to do
CheXagent is weakest on left/right and severity	Test perception alone: mirror-image and heart-size tests
Naive retrieval makes answers worse (RULE)	Short passages, no-harm check, disease-name-only arm
Automatic scores disagree with clinicians	Blind 0-1-2 rating and a constant-answer floor
No reviewed paper retrieves textbook knowledge	BM25 over MedlinePlus and medical textbooks

Research question: can retrieved textbook knowledge make a chest X-ray model answer knowledge questions better?

CheXagent: 77-79% on fine-grained reasoning. RULE: naive retrieval over-trust. Med-Flamingo: clinician rating.

Rules are written before any run, and negative results are reported



219
baseline questions

130
knowledge questions

140
left / right questions

120
mirror-image pairs

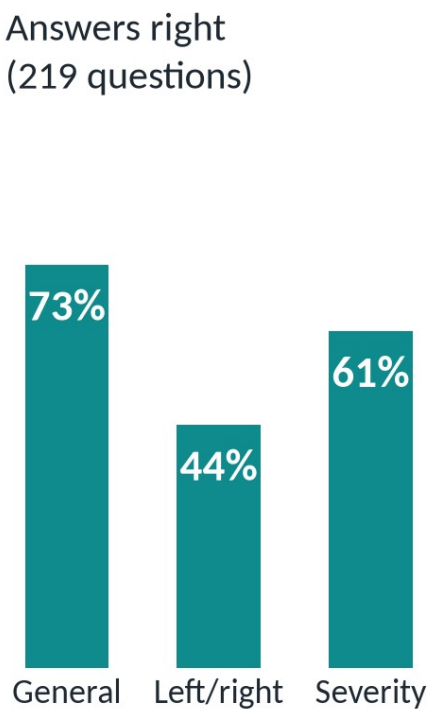
47
heart-size questions

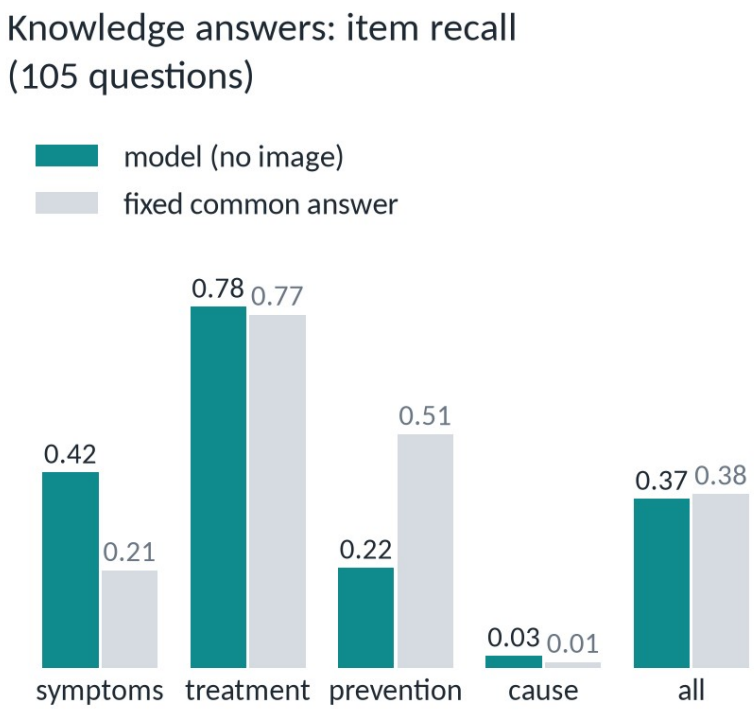
130
images with a disease label

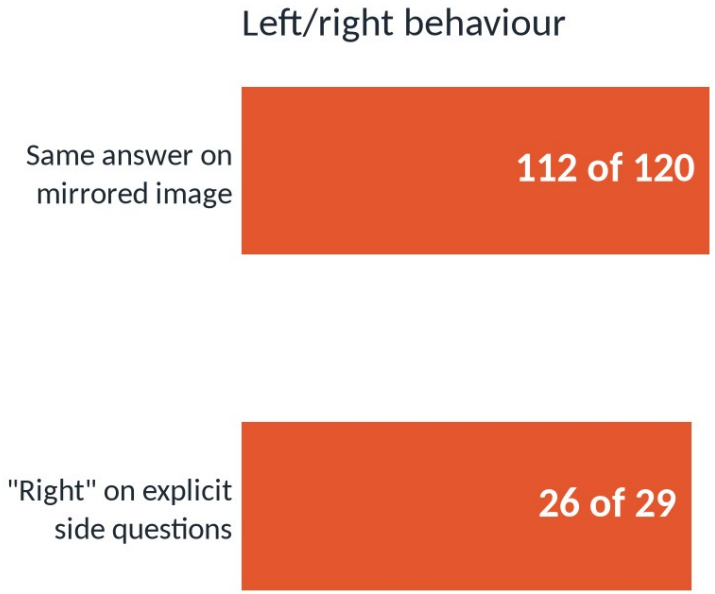
Pre-registration: test sets, scorer and success conditions were frozen first, then amended in writing when needed.

docs/evaluation_protocol.md (6 amendments) and docs/phase2_rag_plan.md. Scorer: item recall v2; floor: most common gold answer.

Our baseline tests: 58% overall, but the left/right and knowledge scores do not hold up





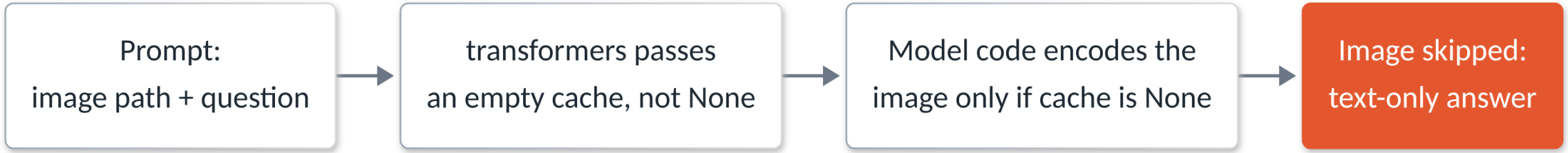


Mirrored image, same answer in 112 of 120 pairs. Knowledge answers: no better than a fixed common answer (0.37 vs 0.38).

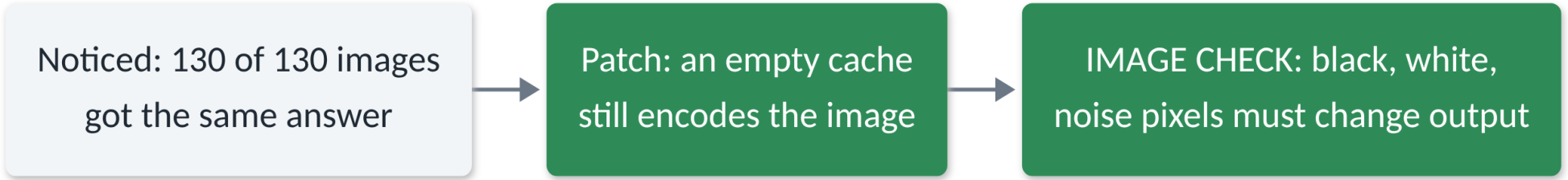
Runs 2-4: 219 baseline questions, 105 knowledge questions, 120 mirror-image pairs. Item recall scorer v2.

Turning point: the image never reached the model, so earlier results are a text-only control

What went wrong



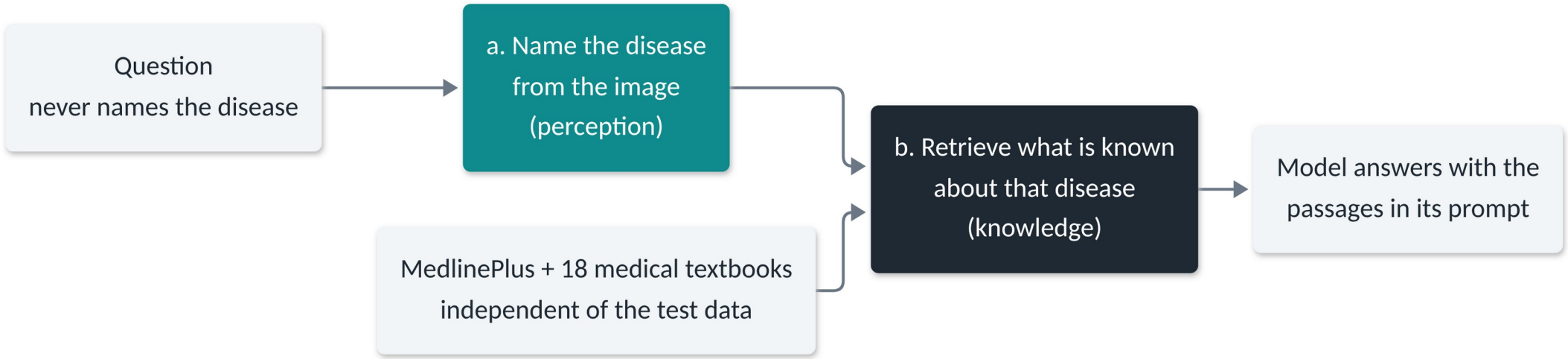
How we found it and fixed it



Still valid: how much of these benchmarks a text-only model solves. Withdrawn: every conclusion about perception.

transformers 4.46.3 passes an empty cache; CheXagent's code only encodes the image when the cache is None. experiment_log.md, entry 17.

Phase 2 design: first name the disease from the image, then retrieve what is known about it



Why two steps: the answer depends only on the disease

Disease on the image	Question about	Gold answer, on every such image
Pneumonia	cause	Bacterial infection
Pneumonia	treatment	Medical therapy, supportive therapy
Cardiomegaly	cause	Hypertension, dilated cardiomyopathy

Corpora: MedlinePlus (US National Library of Medicine, public domain) and medical textbooks (MedRAG); our own BM25 search.

Six arms separate what retrieval adds from what knowing the disease adds

Arm	Disease named	Retrieved text	Role
A1	none	none	fair baseline, same answer style
A2	none	from the question alone	oracle subset: why the disease matters
A3	true disease (oracle)	none	upper bound: value of knowing the disease
A4	true disease (oracle)	yes	upper bound: value of retrieval
A5	model's own guess	none	real: the guess alone
A6	model's own guess	yes	the system

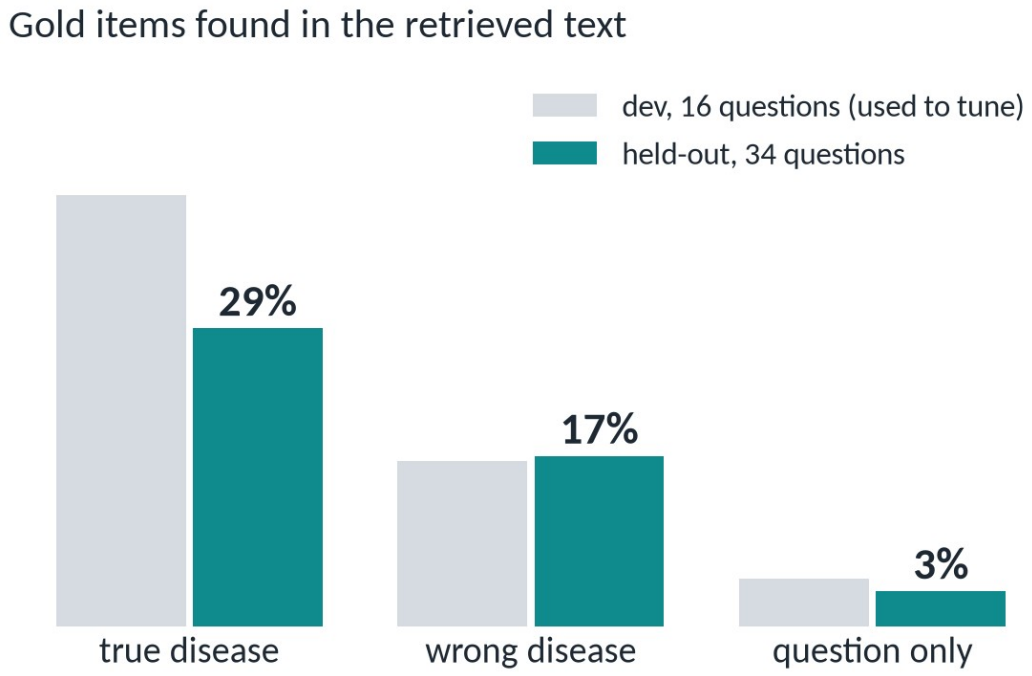
oracle: uses SLAKE's own label, an upper bound, never the result

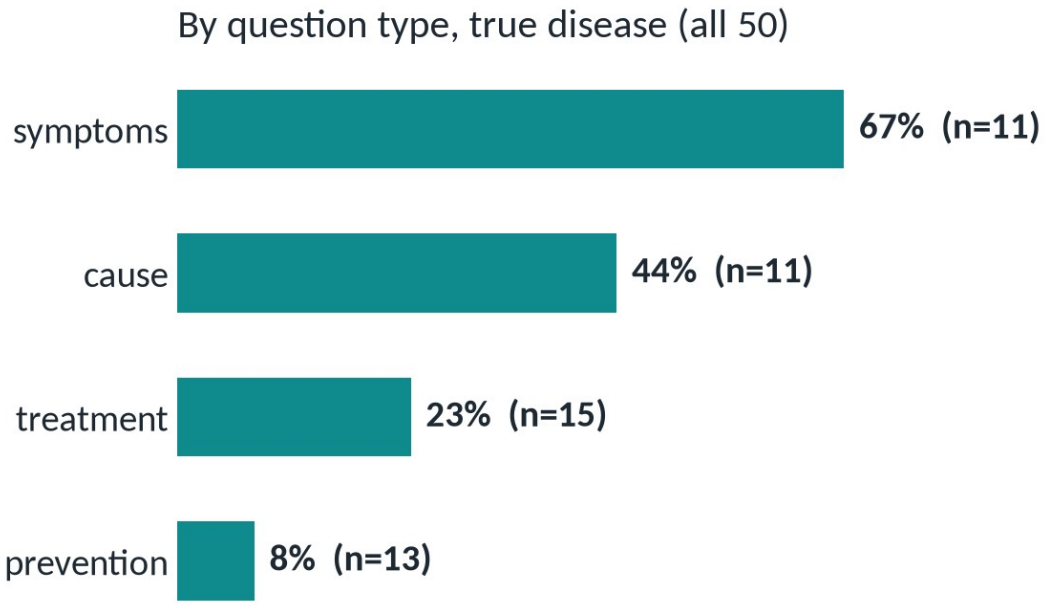
real: uses only what the model outputs

Success needs more than a score gain: the system (A6) must beat both the fair baseline (A1) and the guess alone (A5).

docs/phase2_rag_plan.md and Amendment 5 of the evaluation protocol.

Retrieval check, run without any model: the corpus holds the answer mainly when the disease is right





Held-out: 29% of gold items found with the true disease, 17% with a wrong disease, 3% from the question alone.

Strict coverage: gold item words inside one sentence of the retrieved text. 50 oracle-evaluable questions; setting frozen on 16 dev questions.

References

[1] Chen Z., Varma M., Delbrouck J.-B., et al. CheXagent: Towards a Foundation Model for Chest X-Ray Interpretation. arXiv:2401.12208, 2024.

[2] Xia P., Zhu K., Li H., et al. RULE: Reliable Multimodal RAG for Factuality in Medical Vision Language Models. EMNLP, 2024.

[3] Moor M., Huang Q., Wu S., et al. Med-Flamingo: a Multimodal Medical Few-shot Learner. ML4H, PMLR, 2023.

[4] Eslami S., Meinel C., de Melo G. PubMedCLIP: How Much Does CLIP Benefit VQA in the Medical Domain? Findings of EACL, 2023.

[5] Li C., Wong C., Zhang S., et al. LLaVA-Med: Training a Large Language-and-Vision Assistant for Biomedicine in One Day. NeurIPS D&B, 2023.

[6] Zhang X., Wu C., Zhao Z., et al. PMC-VQA: Visual Instruction Tuning for Medical Visual Question Answering. arXiv:2305.10415, 2023.

[7] Liu B., Zhan L.-M., Xu L., et al. SLAKE: A Semantically-Labeled Knowledge-Enhanced Dataset for Medical VQA. IEEE ISBI, 2021.

[8] Lau J. J., Gayen S., Ben Abacha A., Demner-Fushman D. A dataset of clinically generated visual questions and answers about radiology images. Scientific Data, 2018.

[9] Xiong G., Jin Q., Lu Z., Zhang A. Benchmarking Retrieval-Augmented Generation for Medicine (MedRAG). Findings of ACL, 2024.

[10] Robertson S., Zaragoza H. The Probabilistic Relevance Framework: BM25 and Beyond. Foundations and Trends in IR, 2009.

[11] Haggstrom M. Normal posteroanterior (PA) chest radiograph (X-ray). Wikimedia Commons, CC0.